Agricultural spatial weights Methodology (per NUTS-2 and CORINE agricultural class)

1 Agricultural downscaling framework

1.1 Purpose

CAMS and related atmospheric emission inventories often provide agricultural emissions with spatial resolution that is coarse relative to land-surface processes, or they are aligned to administrative regions. For analysis, comparison with observations, or coupling to land-surface models, one needs a statistical downscaling rule: how to split a regional total for the agricultural sector across finer units that can be mapped to land cover.

1.2 Goal

For each pollutant, assign non-negative weights over CORINE Land Cover agricultural classes within each NUTS-2 region, such that weights sum to unity over classes. CAMS GNFR totals enter the construction through source-specific factors α (mass shares), as described below.

1.3 Methodology

CAMS-REG-ANT distinguishes two agricultural GNFR sectors: **K_AgriLivestock** and **L_AgriOther**. Each aggregates several underlying **emission processes** (enteric fermentation, manure management, fertilized soils, etc.). Table 1 gives an indicative GNFR-to-IPCC mapping for context.

Table 1: Indicative GNFR to IPCC (1996) sector codes

GNFR	IPCC code	Description
L_AgriOther	4A	Enteric fermentation
L_AgriOther	4C	Rice cultivation
L_AgriOther	4D	Soil emissions
L_AgriOther	4F	Biomass burning
K_AgriLivestock	4B	Manure management

See [1] and [2].

To build weights, we first list which **process sources** contribute to each pollutant and whether they sit in GNFR K or L. Table 3 summarizes that mapping. We then combine per-source spatial scores with pollutant-specific European mass fractions $\alpha_{\text{poll}}^{\text{source}}$ between those sources.

Per-source vs. per-pollutant relevance. The spatial factors μ and ρ are defined **per emission source** (e.g. enteric vs. manure management), not as one separate relevance field per pollutant. A given source may influence several pollutants; the pollutant-specific split of mass is carried by α. The implementation will evolve toward a full set of source-specific μ; until then, only a subset of sources has an explicit μ (see Section 2).

Table 2: Agricultural process sources by pollutant and GNFR sector (K/L)

Pollutant	Source	CAMS sector	Reference
CH_4	Enteric fermentation	L	[3]
	Manure management	K	[3]
	Rice paddies	L	[3]
NH_3	Synthetic fertilizer application	L	[3]
	Manure management	K	[3]
	Livestock housing	K	[3]
NO_x	Fertilized soil	L	[4]
	Manure management	K	[4]
	Biomass burning	L	[4]
CO	Biomass burning	L	[5]
CO_2	Biomass burning	L	[5]
	Agricultural soils	L	[6]
	Soil liming	L	[6]
NMVOC	Agricultural soils	L	[7]
	Manure management	L	[7]
	Biomass burning	L	[8]
PM_{10}, $PM_{2.5}$	Biomass burning	L	[9]

(1) Agricultural machinery and pesticides are excluded here: they map to GNFR I (off-road) and E (solvents), not K/L. **(2)** SO_2 from agriculture is negligible for this work (Table 4). **(3)** CAMS carries primary PM only; secondary aerosol is not allocated here, hence biomass burning as the main agricultural PM source in the table.

For each process source, we define a score per CORINE class c and NUTS-2 region n, denoted $s_{n,c}^{\text{source}}$. Weighted by the importance of that source for pollutant p, α_p^{source}, we obtain the combined score $S_{n,c}^p$:

$$S_{n,c}^p = \sum_{\text{source}} s_{n,c}^{\text{source}} \cdot \alpha_p^{\text{source}}. \tag{1}$$

Weights over agricultural classes within NUTS-2 n are obtained by normalizing:

$$w_{n,c}^p = \frac{S_{n,c}^p}{\sum_c S_{n,c}^p}. \tag{2}$$

1.4 Defining scores: class extent and relevance

The product score for a given source uses two ingredients:

- $n_{n,c}$, the **class extent**: how much of CORINE class c lies inside NUTS-2 region n (Section 1.4.2);
- $\rho_{n,c}^{\text{source}} \in [0,1]$, the **relevance** of that source for (n,c), derived from a raw proxy μ (Section 1.4.1).

$$s_{n,c}^{\text{source}} = n_{n,c} \cdot \rho_{n,c}^{\text{source}}. \tag{3}$$

1.4.1 Raw proxy μ and relevance ρ

For each emission source and each pair (n,c), $\mu_{n,c}^{\text{source}}$ is a **non-negative raw spatial proxy**: a quantity that ranks how strongly that source is associated with class c in region n before normalization. The definition is source-specific (grazing activity, fertilizer proxy, manure activity, etc.); concrete choices for the first implemented sources are given in Section 2. Values may be

filled at NUTS-2 × CLC resolution from survey or inventory data, with hierarchical fallbacks when data are sparse (e.g. regional, then country, then pan-European means).

Relevance is obtained by scaling μ to $[0, 1]$ within each country so that the largest μ in that country maps to 1:

$$\rho_{n,c}^{\text{source}} = \text{clip}_{[0,1]}\left(\frac{\mu_{n,c}^{\text{source}}}{\max_{(n',c')\in\mathcal{K}(n)} \mu_{n',c'}^{\text{source}}}\right), \tag{4}$$

where $\mathcal{K}(n)$ is the set of all (NUTS-2, CLC) pairs in the **same country** as region n (same ISO member state). Ties and zero maxima are handled by clipping and the convention that undefined ratios are set to zero.

1.4.2 Class extent $n_{n,c}$

Let the CORINE Land Cover raster be aligned to a fixed grid. For each NUTS-2 polygon n, $n_{n,c}$ is the **number of raster pixels** whose class equals c inside that polygon (equivalently: class c extent in pixel units, or area in pixel area × count). Zonal aggregation yields a histogram $(n_{n,c})_c$ over agricultural classes of interest; only classes present in n contribute positively to downstream sums.

1.5 The α factor

For pollutant p, α_p^{source} is the fraction of **European** agricultural emissions (GNFR K+L combined, or the subset of sources retained in the model) attributed to that process source. Sources included for p are mutually exclusive for budgeting purposes, and their α values sum to unity:

$$\sum_{\text{source}\in p} \alpha_p^{\text{source}} = 1. \tag{5}$$

Example for NMVOC split between agricultural soils and manure management (illustrative):

$$\alpha_p^{\text{Agri. soils}} = \frac{E_p^{\text{Agri. soils}}}{E_p^{\text{Agri. soils}} + E_p^{\text{Manure manag.}}}. \tag{6}$$

Table for alpha values

Table 3: Emissions - in Gg - by pollutant and agricultural source and α values

Pollutant	Source	Emissions	α
CH_4	Enteric fermentation	6575.21	0.790
	Manure management	1645.17	0.198
	Rice paddies	98.32	0.012
NH_3	Synthetic fertilizer application	1749	0.541
	Manure management + Livestock housing	1481	0.459
NO_x	Fertilized soil	638	0.936
	Manure management	37.6	0.055
	Biomass burning	5.8	0.009
CO_2	Carbon fertlizer	4704.57	0.468
	Soil liming	5346.93	0.532
NMVOC	Agricultural soils	330.96	0.239
	Manure management	1052.35	0.760
	Biomass burning	2.04	0.001

(1) For pollutant : CO, CO_2,CH_4 and NMVOC we use the emissions provided by the European Environment Agency (EEA) for the year 2020 see [11]. For NH_3 and NO_x we use the values provided by the 2023 Guidebook [10]. **(2)** No need to compute α ratio for Particulate matter and carbon monoxide as they only have one source of agricultural emissions.

2 Per-source relevance factors (μ, ρ)

This section records how μ is built for each process source once implemented. Sources not yet tied to a μ pipeline are marked for future work.

2.1 Enteric fermentation

Relevant pollutants: CH_4 (GNFR L).

Grazing-based relevance. Enteric CH_4 is linked to ruminant husbandry on grassland. The raw proxy $\mu_{n,c}^{\text{enteric}}$ uses a **grazing activity indicator** aggregated from LUCAS (and related) survey flags at NUTS-2 $\times$ CLC resolution, then passed through the normalization in Section 1.4.1. Implementation column names include `r_ch4` /

2.2 Manure management

Relevant pollutants: CH_4 (manure), NH_3, NO_x, NMVOC (as in Table 3).

Manure-based relevance. The raw proxy $\mu_{n,c}^{\text{manure}}$ uses a **manure management indicator** (survey-based flags for storage, spreading, housing, etc.) at NUTS-2 $\times$ CLC, then normalized as in Section 1.4.1.

2.3 Livestock housing

Relevant pollutants: NH_3 (GNFR K).

Source-specific μ and ρ: To be specified when a dedicated housing relevance factor is implemented, or merged explicitly with the manure pipeline above.

2.4 Rice paddies

Relevant pollutants: CH_4 (GNFR L).

Source-specific μ and ρ: To be specified when a dedicated rice relevance factor is implemented.

2.5 Synthetic fertilizer and fertilized soil

Relevant pollutants: NH_3 (synthetic application), NO_x (fertilized soil).

Synthetic nitrogen proxy. For soil-related emissions tied to mineral fertilizer, $\mu_{n,c}^{\text{synth}}$ uses a **synthetic nitrogen rate proxy** (or equivalent agronomic surrogate) at NUTS-2 × CLC, then normalized as in Section 1.4.1. Implementation: `r_synthetic` / `relevance_synth_n`.

2.6 Biomass burning

Relevant pollutants: CO, CO_2, NO_x, PM (Table 3).

Source-specific μ and ρ: To be specified when a dedicated biomass-burning relevance factor is implemented.

2.7 Agricultural soils and liming

Relevant pollutants: CO_2, NMVOC (Table 3).

Source-specific μ and ρ: To be specified when dedicated soil / liming relevance factors are implemented.

3 Annex

Table 4: CAMS-REG-ANT v8.1: total emissions by pollutant for GNFR K (AgriLivestock) and L (AgriOther), summing area and point sources. Units: kg yr^{-1}.

Pollutant	K (AgriLivestock)	L (AgriOther)
CH_4	1.529×10^{10}	6.434×10^{8}
CO	0	1.413×10^{9}
NH_3	3.172×10^{9}	3.243×10^{9}
NMVOC	2.612×10^{9}	7.561×10^{8}
NO_x	7.319×10^{6}	1.071×10^{8}
PM_{10}	2.745×10^{8}	4.742×10^{8}
$PM_{2.5}$	5.542×10^{7}	1.483×10^{8}
SO_x	0	6.351×10^{6}
CO_2 (fossil)	0	1.099×10^{10}
CO_2 (biofuel)	0	2.490×10^{10}

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